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TEXAS
The University of Texas at Austin

EXAMINING THE ROLE OF BROKERING FREQUENCY IN PREDICTING LONG-TERM ACADEMIC OUTCOMES OF CHINESE AMERICAN ADOLESCENTS

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INTRODUCTION

Introduction-Background

- **33%** of Chinese immigrant households **lack family members** aged 14 or over who are fully proficient in English (Dietrich & Hernandez, 2022).
- **Over than 60%** of **Chinese American youth report** engaging in **language brokering** for their parents since early adolescence (Chao, 2006).
- Language brokering is a **unique developmental task** faced only by immigrant-origin youth, can indicated their level of adaptation (García Coll et al., 1996).
- Language brokering frequency have important influence for adolescent **academic outcomes** (Shen et al., 2022).

Introduction-Language Brokering Frequency & Academic Performance

- **Positive effects**: Improved cognitive skills & expand vocabulary (Kam & Lazarevic, 2014; Borrero, 2015).
- **Negative effects**: Reduced school work time, lower self-concept (Niehaus & Kumpiene, 2014; Jones & Trickett, 2005).
- Meta-analysis: **no significant linear association** between language brokering frequency and academic outcomes (Shen et al., 2022).

1. Non-linear relationship?
2. Different effects on distinct subjects?

Introduction-Non-linear relationship

1. Studies on adolescent work

- Steinberg & Dornbusch (1991); Hwang & Domina (2017):
  - Intensive work (>20hr) → negative (lower academic investment and performance)
  - Moderate work hours (1–20hr/<5hr) → no difference; positive

2. Language brokering on other outcomes

- Kam et al. (2017)
  - Infrequent and occasional brokering → lower ethnic discrimination.
  - Frequently brokering → higher ethnic discrimination

Optimal at the Moderate Level:
Potential for Curvilinear Effects

Introduction-Language Brokering on Different Subjects

1. **English language arts**: heavily depend on vocabulary and semantic processing, may benefit more directly (Martinez et al., 2009).

2. **Mathematics:** less language-dependent, may exhibit limited benefit (Vukovic & Lesaux, 2013)

3. **GPA:** mixed findings but can provide a comprehensive perspective of general academic development.

Current study

RQ: How language brokering frequency predicts academic performance & the trajectory of academic performance?

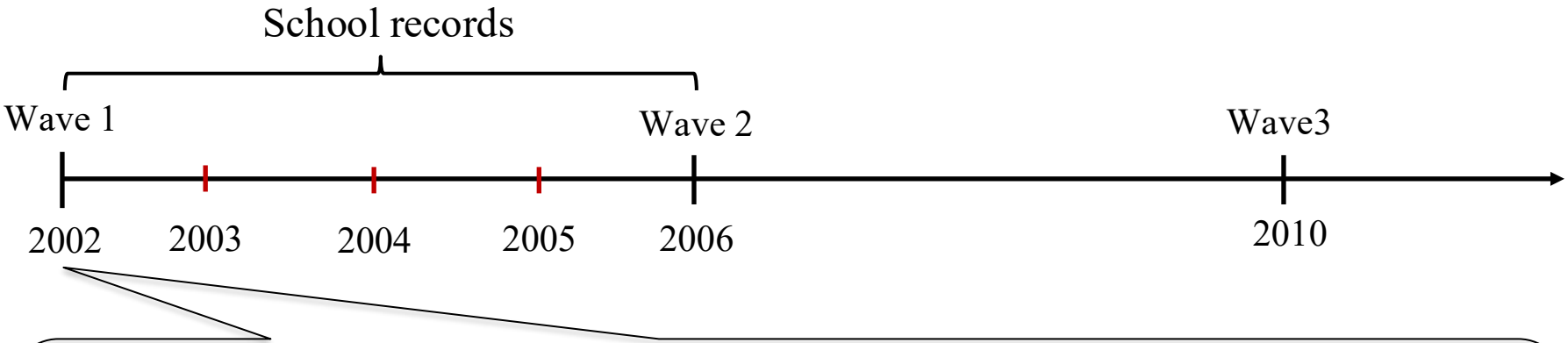
Hypotheses 1: Moderate brokering frequency will be linked to Chinese American adolescents' higher initial academic achievement & more positive academic changes

Hypotheses 2: Moderate brokering frequency have positive influence on English Language Art score and GPA, but not on Math score.

METHODS

Methods-Participants

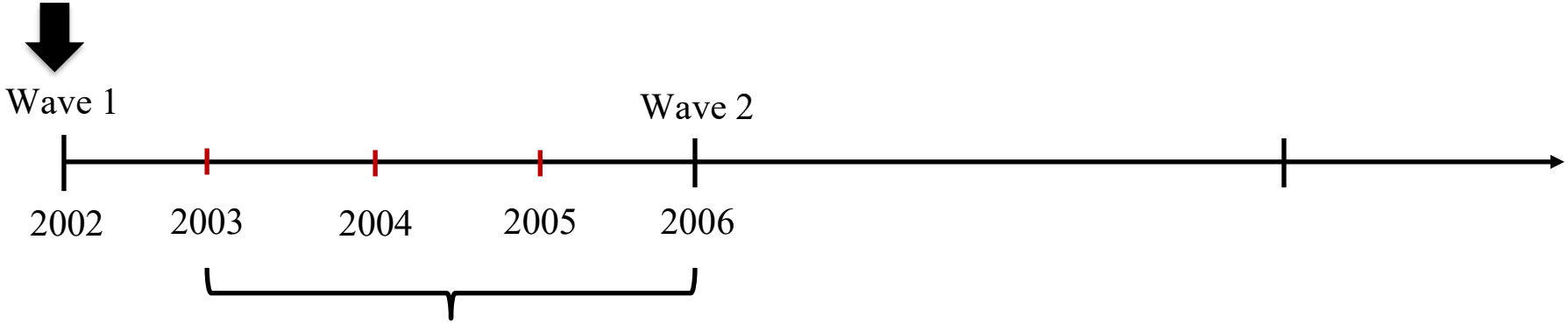
A three-wave longitudinal data (PI Kim: 1R03HD051629-01A2):



- 444 Chinese American adolescents (53.6% female) and their parents in Northern California
- Adolescents' age ranged from 12 to 15 (M = 13.04; SD = 0.73).
- Most adolescents (n = 334, 75.2%) were born in the U.S., most of the parents were foreign-born.
- Most adolescents reported speaking Chinese at home (n = 409, 92.1%) .
- The median income was in the range of $30,001 to $45,000 at Wave 1.

Methods-Participants

Self-reported language brokering frequency



School records:

- English language art (ELA)
- Math
- GPA

Methods-Measures

Language Brokering frequency:

- How often your had translated from English to Chinese for you mother/father?
- 1=Never, 2=A few times a year, 3=A few times a month, 4= A few times a week, 5=Daily

Academic Outcomes:

- GPA: 4.0 Scale
- ELA & Math:100 (far below the basic level) to 500 (advanced level)

Covariates:

- Adolescents' age, gender, nativity, and parental education level
- Contextual stresses: Economic Stress, Discrimination, Acculturative family conflict

Methods-Analyses

Step 1: Growth Curve Modeling for GPA, math, ELA scores

Step 2-4: OLS regressions

- Brokering frequency to initial level (intercept) and trajectories (slope) of academic outcomes
- + quadratic term of language brokering frequency
- + contextual stress

- *Maximum Likelihood Parameter Estimates.*
- *Full Information Maximum Likelihood (FIML) for missing data.*

RESULTS

1. The change of academic performance over the year

GPA:

- Positive intercept (b = 3.205, SE = 0.034, p <.001; σ^2 = 0.412, SE = 0.050, p < .001) & Significant **decrease** across time (b = -0.029, SE = 0.010, p = .004; σ^2 = 0.015, SE = 0.004, p = .001)

Math:

- Positive intercept (b = 338.257, SE = 11.256, p <.001; σ^2 = 39632.176, SE = 12594.970, p = .002) & Significant **decrease** across (b = -36.031, SE = 5.431, p <.001; σ^2 = 8250.142, SE = 2683.304, p = .002)

ELA:

- Positive intercept (b = 354.105, SE = 2.287, p <.001; σ^2 = 1861.279, SE = 148.958, p <.001) &Significant **increase** across time (b = 6.308, SE = 0.553, p <.001; σ^2 = 26.511, SE = 13.147, p = .044).

2. Language brokering to GPA

Table 2

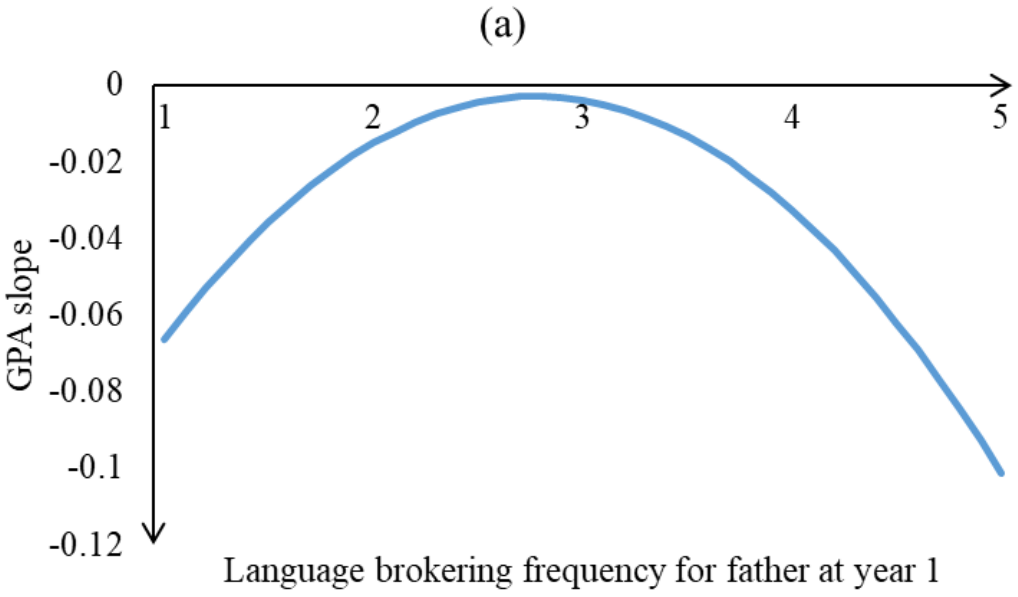
Growth curve model results for GPA

	Model 1		Model 2		Model 3	
	Intercept *b* (*SE*)	Slope *b* (*SE*)	Intercept *b* (*SE*)	Slope *b* (*SE*)	Intercept *b* (*SE*)	Slope *b* (*SE*)
Mother model						
M-LB frequency	0.034 (0.040)	-0.011 (0.013)	0.036 (0.040)	-0.013 (0.013)	0.050 (0.039)	-0.014 (0.013)
M-LB frequency^2			0.015 (0.037)	-0.014 (0.011)	0.019 (0.035)	-0.015 (0.011)
Economic stress					**-0.105* (0.046)**	0.007 (0.013)
Discrimination					**-0.171* (0.071)**	0.018 (0.022)
Acculturative family conflict with mother					-0.046 (0.042)	-0.016 (0.012)
CFI	0.982		0.978		0.977	
RMSEA	0.046		0.049		0.044	
Father model						
F-LB frequency	-0.003 (0.034)	-0.003 (0.011)	-0.003 (0.034)	-0.004 (0.010)	-0.003 (0.033)	-0.003 (0.010)
F-LB frequency^2			-0.010 (0.025)	**-0.019* (0.008)**	-0.001 (0.024)	**-0.020* (0.008)**
Economic stress					**-0.099* (0.045)**	0.005 (0.012)
Discrimination					**-0.164* (0.071)**	0.015 (0.022)
Acculturative family conflict with father					-0.024 (0.045)	-0.009 (0.012)
CFI	0.98		0.98		0.974	
RMSEA	0.049		0.047		0.048	

Note. F = father; M = mother; LB = language brokering. Adolescent age, gender, nativity, study hours, and maternal/paternal education were included as covariates. Significant findings are bold.

*p < .05, ** p < .01.

2. Language brokering to GPA



Youth who reported moderate language brokering frequency (i.e., A few times a month) for fathers at Year 1 experienced a flatter decrease in GPA

Note. 1=Never, 2=A few times a year, 3=A few times a month, 4=A few times a week, 5=Daily.

3. Language brokering to math

Table 3

Growth curve model results for math

	Model 1		Model 2		Model 3	
	Intercept *b* (*SE*)	Slope *b* (*SE*)	Intercept *b* (*SE*)	Slope *b* (*SE*)	Intercept *b* (*SE*)	Slope *b* (*SE*)
Mother model						
M-LB frequency	5.362 (4.020)	-0.118 (1.190)	5.074 (4.209)	-0.206 (1.243)	6.435 (4.170)	-0.044 (1.242)
M-LB frequency^2			-1.533 (3.603)	-0.572 (0.995)	-1.652 (3.585)	-0.596 (0.999)
Economic stress					-4.897 (3.378)	0.452 (0.979)
Discrimination					-10.523 (5.570)	2.278 (1.935)
Acculturative family conflict with mother					**-8.055** (3.081)**	-1.848 (1.044)
CFI	0.932		0.931		0.937	
RMSEA	0.091		0.088		0.074	
Father model						
F-LB frequency	-0.873 (3.462)	0.164 (0.961)	-1.062 (3.411)	0.147 (0.965)	-0.586 (3.356)	0.270 (0.964)
F-LB frequency^2			-4.663 (3.021)	-1.033 (0.849)	-4.191 (3.020)	-0.990 (0.852)
Economic stress					-3.763 (3.464)	0.357 (0.957)
Discrimination					-7.056 (5.704)	1.885 (1.998)
Acculturative family conflict with father					**-6.997* (2.941）**	-1.036 (1.029)
CFI	0.931		0.932		0.939	
RMSEA	0.093		0.087		0.072	

Note. F = father; M = mother; LB = language brokering. Adolescent age, gender, nativity, study hours, and maternal/paternal education were included as covariates. Significant findings are bold.

*p < .05, ** p < .01.

4. Language brokering to ELA

Table 4

Growth curve model results for English language art

	Model 1		Model 2		Model 3	
	Intercept *b* (*SE*)	Slope *b* (*SE*)	Intercept *b* (*SE*)	Slope *b* (*SE*)	Intercept *b* (*SE*)	Slope *b* (*SE*)
Mother model						
M-LB frequency	1.810 (2.852)	0.525 (0.722)	0.839 (2.749)	0.547 (0.731)	1.793 (2.745)	0.557 (0.726)
M-LB frequency^2			**-4.925* (2.294)**	0.043 (0.575)	**-4.855* (2.181)**	0.005 (0.566)
Economic stress					**-5.793* (2.744)**	-0.193 (0.598)
Discrimination					-5.039 (4.655)	1.388 (1.258)
Acculturative family conflict with mother					-4.593 (2.562)	**-1.508* (0.593)**
CFI	0.943		0.944		0.946	
RMSEA	0.104		0.098		0.085	
Father model						
F-LB frequency	-0.893 (2.406)	-0.500 (0.548)	-1.063 (2.346)	-0.480 (0.549)	-0.852 (2.361)	-0.386 (0.551)
F-LB frequency^2			**-4.951** (1.734)**	0.071 (0.457)	**-4.589** (1.753)**	0.050 (0.45)
Economic stress					-5.323 (2.818)	-0.211 (0.608)
Discrimination					-3.189 (4.704)	1.084 (1.25)
Acculturative family conflict with father					-2.702 (2.707)	-0.614 (0.593)
CFI	0.945		0.946		0.947	
RMSEA	0.103		0.097		0.084	

Note. F = father; M = mother; LB = language brokering. Adolescent age, gender, nativity, study hours, and maternal/paternal education were included as covariates. Significant findings are bold.

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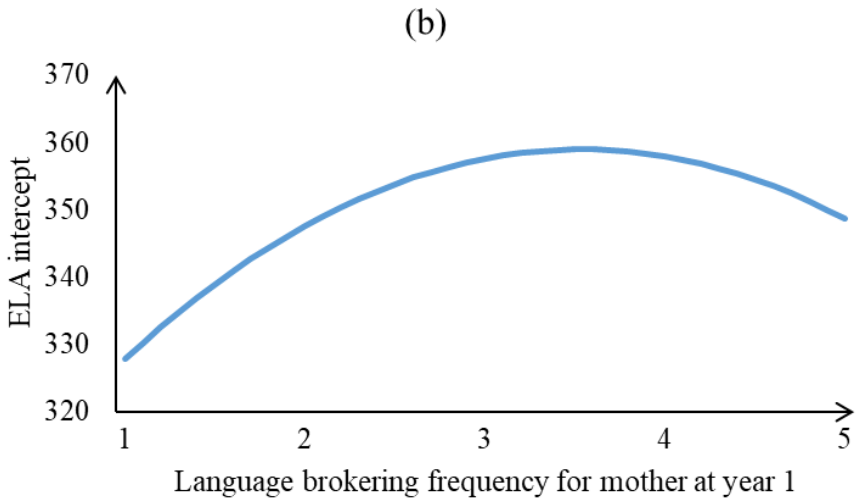
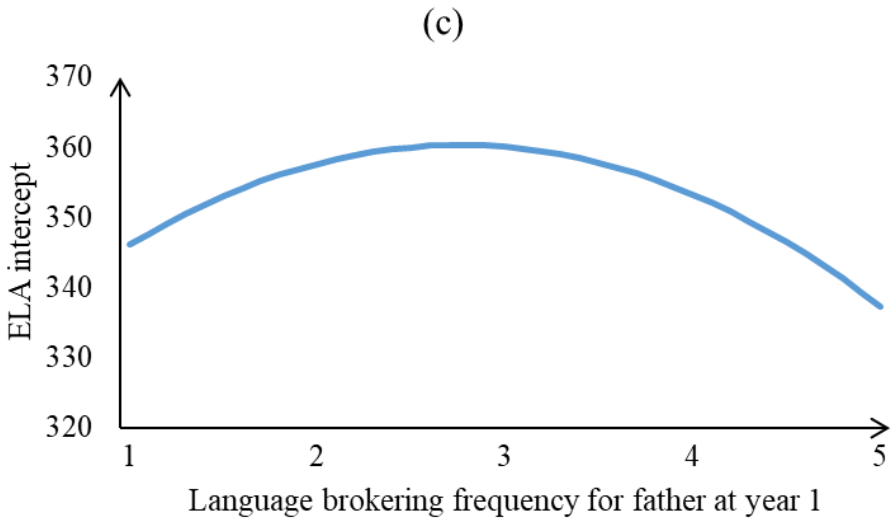
4. Language brokering to ELA





Note. 1=Never, 2=A few times a year, 3=A few times a month, 4= A few times a week, 5=Daily.

Youth who reported moderate language brokering frequency (i.e., a few times a month) had the highest English language art scores

4. Language brokering to ELA

Table 4

Growth curve model results for English language art

	Model 1		Model 2		Model 3	
	Intercept *b* (*SE*)	Slope *b* (*SE*)	Intercept *b* (*SE*)	Slope *b* (*SE*)	Intercept *b* (*SE*)	Slope *b* (*SE*)
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Note. F = father; M = mother; LB = language brokering. Adolescent age, gender, nativity, study hours, and maternal/paternal education were included as covariates. Significant findings are bold.

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Conclusion

1. Moderate brokering benefit GPA/ELA:

- Moderate language brokering frequency (i.e., a few times a month) for fathers can flatten decrease in GPA across years.
- Moderate language brokering frequency (a few times a month) can help improve the initial level English language art score.

2. Domain-specific effects of language brokering: ELA > Math.

3. Contextual stressors have independent negative impacts.

Contributions & Implication

Past studies:

- Exploration of moderators across individual (e.g., self-efficacy, obligation), family (e.g., parental warmth, hostility), and societal (e.g., discrimination) levels

This study:

- Consider the nonlinear associations
- Distinguished academic domains
- Include contextual stressors

Implication:

- Understanding the dual-edged nature of language brokering can inform support strategies
- Parents recognize the burdens and benefits of brokering to promote balanced expectations
- Schools could offer resources for students heavily engaged in brokering

THANKS!

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Table 1

Descriptive Statistics and Correlation of Study Variables

Variable	*M*	*SD*	1	2	3	4	5	6	7	8	9	10
1. Nativity (0 = Mexico; 1 = US)	0.754	—	—									
2. Gender (0 = male; 1 = female)	0.54	—	.007	—								
3. Age	13.042	0.728	.101*	.006	—							
4. Paternal education	5.927	1.817	–.047	.016	–.096	—						
5. Maternal education	5.861	1.718	–.142**	–.023	–.075	.621***	—					
6. Study hours	3.151	1.612	–.056	.199***	–.068	.074	.039	—				
7. M-LB frequency	3.351	0.946	.006	–.110	.032	–.113	–.178**	–.040	—			
8. F-LB frequency	2.854	1.129	.041	–.022	.079	–.233***	–.036	.036	.490***	—		
9. Economic stress	1.594	0.836	–.026	–.016	.002	–.082	–.077	–.035	.078	.034	—	
10. Discrimination	1.724	0.503	.018	–.064	.062	–.191***	–.111*	–.093	.113	.042	.190***	—
11. M-Acculturative family conflict	2.657	0.855	.040	–.009	.039	–.154**	–.059	–.010	.060	.132*	.172***	.309***
12. F-Acculturative family conflict	2.543	0.864	.087	–.048	.046	–.126*	–.063	–.056	.068	.130*	.181***	.281***
13. GPA year 1	3.174	0.744	–.076	.314***	–.172**	.139*	.153**	.159**	.049	–.012	–.121*	–.155**
14. GPA year 2	3.164	0.687	–.012	.300***	–.075	.252***	.193***	.187***	–.064	–.070	–.153**	–.200***
15. GPA year 3	3.136	0.644	–.052	.249***	.026	.151**	.144**	.134**	–.004	–.037	–.196***	–.180***
16. GPA year 4	3.157	0.582	–.038	.284***	.006	.138*	.129*	.114*	–.065	–.084	–.130*	–.160**
17. Math year 1	378.42	59.324	.052	.106*	.006	.143**	.075	.189***	.060	–.041	–.085	–.135**
18. Math year 2	387.4	68.393	.107*	.048	–.085	.183***	.068	.105*	.047	–.079	–.118*	–.121*
19. Math year 3	365.77	65.314	.069	.104*	–.067	.087	.003	.129**	.068	–.037	–.089	–.108*
20. Math year 4	359.6	65.88	.033	.064	–.173***	.111*	.027	.062	.040	–.079	–.035	–.065
21. ELA year 1	350.7	47.128	–.232***	.122*	–.020	.248***	.251***	.129**	–.013	–.061	–.149**	–.160***
22. ELA year 2	366.01	48.265	–.252***	.171***	.075	.203***	.194***	.090	.014	–.097	–.096	–.078
23. ELA year 3	369.43	44.889	–.198***	.183***	–.055	.220***	.177***	.122*	.018	–.127*	–.153**	–.121*
24. ELA year 4	369.35	47.284	–.178***	.167***	–.076	.229***	.193***	.154**	–.005	–.128*	–.123*	–.107*

Note. M = mother; F = father; LB = language brokering; ELA = English language art.